Hybrid methods for well-mixed chemical reaction systems

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Outline

1. Model
2. Gillespie's SSA
3. Discrete system to ODE
4. Hybrid method

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Chemical reaction system

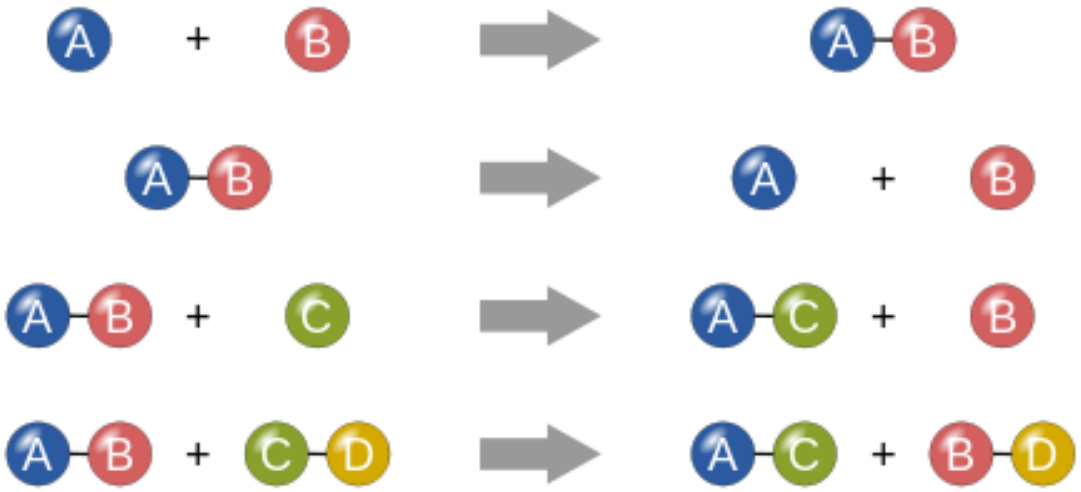

Figure 1: Chemical reaction

Reaction networks

The *network* of r_0 chemical reactions involving s_0 chemical species, $S_1, ..., S_{s_0}$

$$\sum_{i=1}^{s_0} v_{ik} S_i \rightharpoonup \sum_{i=1}^{s_0} v'_{ik} S_i, \qquad k = 1, ..., r_0 \qquad v_{ik}, v'_{ik} \in \mathbb{Z}^+$$

- v_{ik} : the coefficients of ith spiecies in kth reaction before reaction
- v'_{ik} : the coefficients of ith spiecies in kth reaction after reaction

Theorem

Assuming $P\{reaction\ occurs\ in\ (t, t+\triangle t]|F_t)\} = \lambda_k(X(t))\triangle t$*,the state of the system at time t satisfies:*

$$\begin{aligned} X(t) &= X(0) + \sum_k R_k(t)(v'_k - v_k) \\ &= X(0) + \sum_k Y_k(\int_0^t \lambda_k^N(X(s))ds)(v'_k - v_k) \end{aligned} \tag{1}$$

where $R_k(t)$*is the number of times that* k_{th}*reaction occurs before time t,*Y_k *is unit-rate Possion process.*

Proof.

Simple verification using properties of Possion process. □

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SSA pesudocode

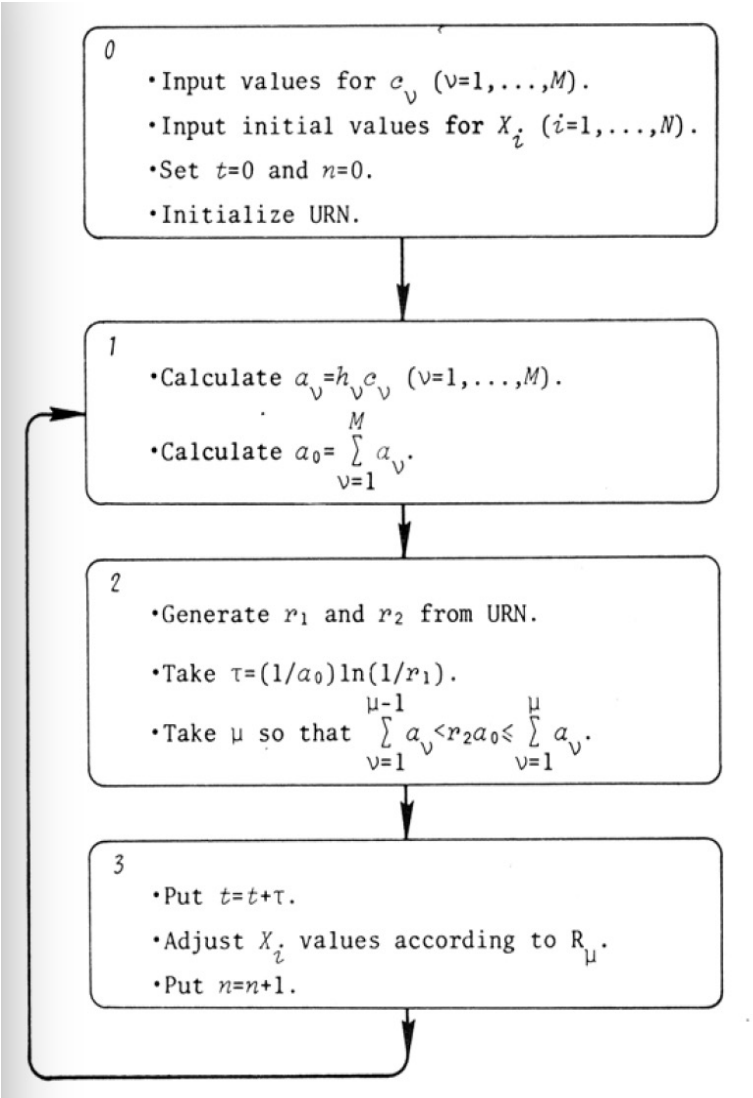

Figure 2: *direct method of SSA*

Theorem

SSA is an exact simulation of (1),in other words,SSA is fully equvalent to (1).

Proof.

SSA is an exact simulation of the following reaction probability density function:

$$P(\mu, \tau)d\tau = a_\tau exp(-\tau \sum_j a_j)d\tau \tag{2}$$

It suffies to show (2) is equivalent to (1),which can found in David F.Anderson,Thomas.G.Kurtz *Continuous time markov chain models for chemical reaction networks* (2011) □

Disadvantage

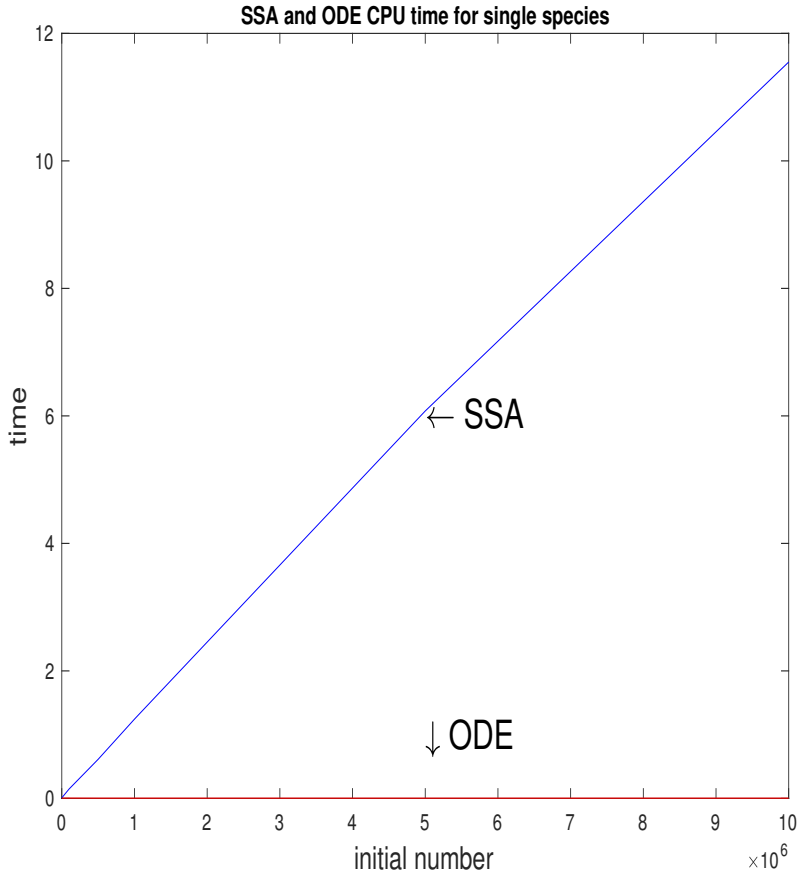

Figure 3: *SSA and ODE computation time of simulating the system to $T = 10s$ with different initial number in a simple system*

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Theorem

In (1),if we replace X(t) by $z(t) = X(t)/N$, the stochastic equation of $z(t)$ can be approximated by an ordinary differential equation:

$$z'(t) = \sum_k \alpha_k \prod_i z_i(t)^{v_{ik}} (v'_k - v_k) \tag{3}$$

Futhermore, Let $z(t)_{sto}$ be the solution of stochastic equation, $z(t)_{ode}$be the solution of coresponding ordinary differential equation, we have:

$$z_{sto}(t) = z_{ode}(t) + O(\frac{1}{\sqrt{N}}) \tag{4}$$

Proof.

For first part:

In general,consider stochastic equation:

$$X_N(t) = X_N(0) + \sum \frac{1}{N} l Y_l(N \int_0^t f_l^N(X_N(s))ds) \tag{5}$$

For large values of N, The Law of Large Number implies:

$$\frac{Y_l(Nu)}{N} \approx u \tag{6}$$

Then we get the coresponding ordinary differential equation:

$$X(t) = X(0) + \int_0^t \sum l f_l(X(s))ds \tag{7}$$

□

Disadvantage

When the population of species is not large, ODE is not a good approximation of stochastic equation.

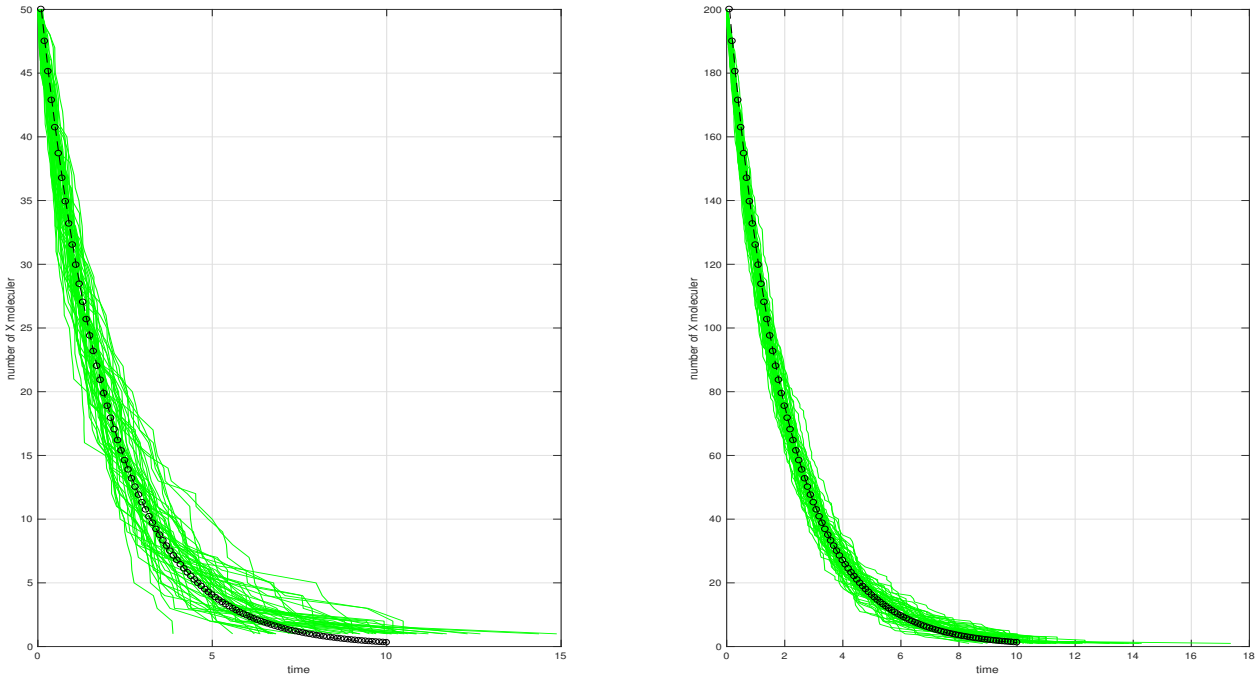

Figure 4: Histogram of 50 SSA simulations up to time T = 10s with X(0) = 50 and 200. The black circle line is the solution of coresponding ODE by Euler method

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Blending functions

- For reactions that involve only 1 species: We can define

$$\beta(x, I_1, I_2) = \begin{cases} 1, & x \leq I_1 \\ \frac{I_2 - x}{I_2 - I_1}, & I_1 < x < I_2 \\ 0, & x \geq I_2 \end{cases} \tag{8}$$

- For reacions that involve multiple species.
  Let S_r be the set of species involved in r-th reaction(Both as reactants and products), we define:

$$\beta_r(X) = 1 - \prod_{n \in S_r} (1 - \beta(X_n, I_1^n, I_2^n))$$

The hybrid scheme

We consider the following equation:

$$X(t) = X(0) + \sum_k Y_k(\int_0^t \beta_r(X(s))\lambda_k([X(s)])ds)(v'_k - v_k) + \sum_k (v'_k - v_k)\int_0^t (1 - \beta_r(X(s))\alpha_k \prod_i X_i(s)^{v_{ik}} ds \tag{9}$$

where $[X(s)]$ is the closet point in lattice $\mathbb{Z}^{s_0}$ to X(s)$\in \mathbb{R}^{s_0}$
Remark :In regions where $\beta_r(X) = 1$the dynamics of the r-th reaction is modelled by the pure SSA, when $\beta_r(X) = 0$, the dynamics is pure ODE, in intermediate regions where $0 < \beta_r(X) < 1$ we abtain a mixture of the two.

Algorithm

Input:
Propensities $\alpha_1, ..., \alpha_{r_0}$;
blending functions $\beta_1, ..., \beta_{r_0}$;
stoichimetric matrix for reactants:$v = (v_1, v_2, ..., v_{r_0})$,for products:$v' = (v'_1, v'_2, ..., v'_{r_0})$
final time of simulation T;
time step for ODE:$\Delta t, \delta t (\delta t > \Delta t)$;
initial state $X(0) \in \mathbb{Z}^{s_0}$

Algorithm

```
Set t = 0;
while t < T do
    if max_j β_j(X(t)) = 0 then
        Simulate ODE (6) with propensity α_i by Euler method up to time t + δt;
        Set t = t + δt;
    else if min_j β_j(X(t)) = 1 then
        Compute λ_0 = Σ_{j=1}^{r_0} λ_j([X(t)]);
        Generate r_1, r_2 from URN Take τ = (1/λ_0) ln(1/r_1);
        Take μ so that Σ_{v=1}^{μ-1} λ_v < r_2 λ_0 < Σ_{v=1}^{μ} λ_v;
        Set X(t + τ) = X(t) + v'_μ − v_μ;
        Set t = t + τ ;
    else
        Compute λ'_0 = Σ_{j=1}^{r_0} β_j(X(t)) λ_j([X(t)]);
        Generate r_1 from URN;
        Take τ = (1/λ'_0) ln(1/r_1);
        if τ < Δ t then
            Generate r_2 from URN;
            Take μ so that Σ_{v=1}^{μ-1} λ_v < r_2 λ_0 < Σ_{v=1}^{μ} λ_v;
            Simulate ODE (6) with propensity (1 − β_j(X)) α_j(X) up to time t + τ and set X(t + τ) = X(t + τ) + v'_μ − v_μ;
            Set t = t + Δt;
        else
            Simulate ODE (6) with propensity (1 − β_j(X)) α_j(X) up to time t + Δt;
            Set t = t + Δt;
        end
    end
end
```

Algorithm 1: Simulation of hybrid model

Numerical investigations:efficiency

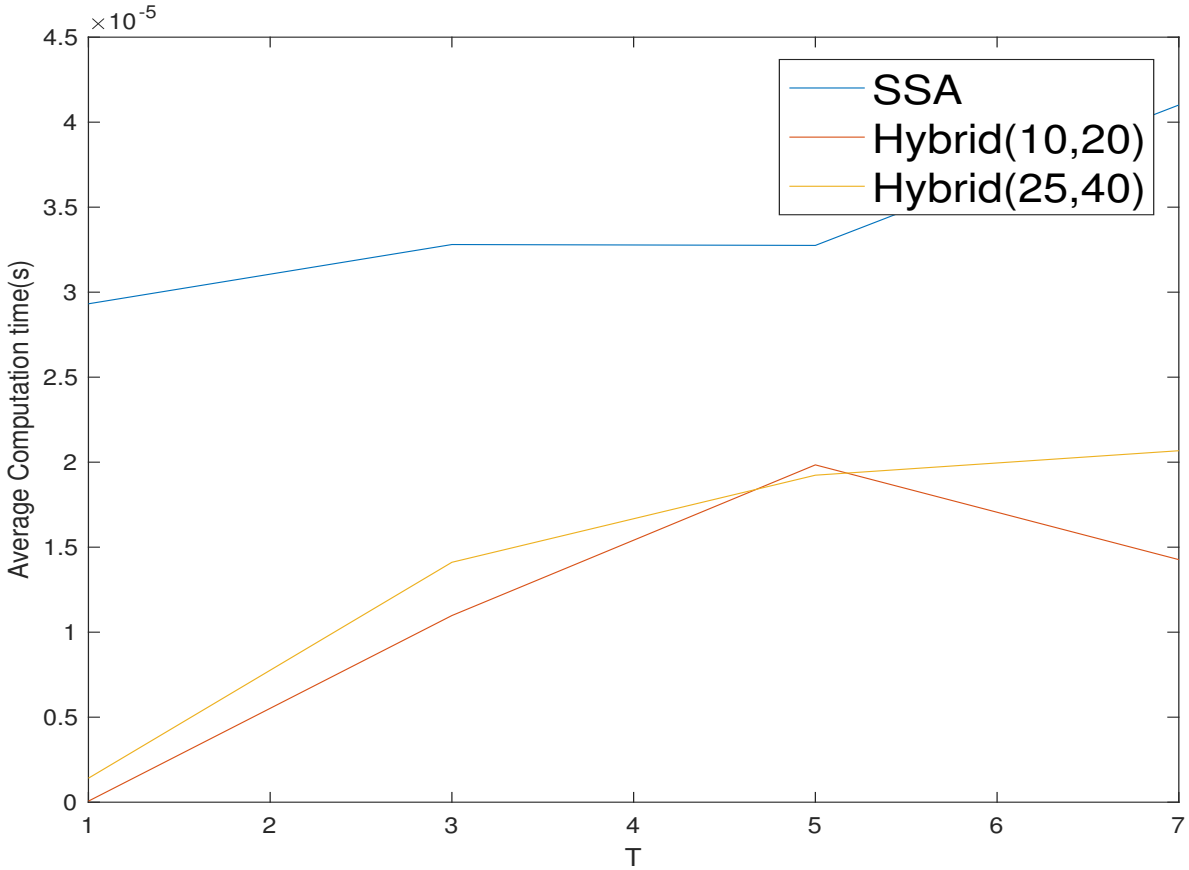

Figure 5: Average computation time as a function of T,X(0) = 60, h = 0.1

Numerical investigations:accuracy

T	SSA	ODE	Hybrid(10,20)
3s	12.000	12.800	11.450
5s	3.000	4.616	2.870
7s	0.800	1.655	1.201

Table 1: Average simulation result of three methods with X(0) = 60,h=0.1

Conclusion

Numerical investigations show that by choosing blending functions and time step of Euler method properly, this hybrid method perform well both in accuracy(compared with pure ODE) and efficiency(compared with pure SSA).

Thank you!